SCHEDULE 17: DEEMED ENERGY CALCULATION - METHODOLOGY

1. Definitions and Interpretation

1.1 Except as otherwise defined herein or in any attachment hereto or as the context may otherwise require, any capitalised term used in this schedule 17 shall have the meaning set forth in the Power Purchase Agreement (**"Agreement"**) to which this schedule 17 is attached.

1.2 Unless otherwise specified, references to "sections" or "appendices" in this schedule 17 are references to sections of, and appendices to, this schedule 17. References to "clauses" and "schedules" in this schedule 17 are references to clauses and schedules of the Agreement.

"Deemed ICOD" means the Initial Commercial Operation Date has deemed to be achieved in accordance with clause 8.3 of the Agreement;[1]

"Deemed Net Electrical Energy Interval" means any interval of 60 minutes where the Offtaker is required to pay Deemed Energy Payments, in accordance with this Agreement.

"Deemed PCOD" means the Project Commercial Operation Date has deemed to be achieved in accordance with clause 8.4 of the Agreement;

"Initial Nominal Installed Power Rating" or "IPNOM" means the sum of the nameplate rating at 50°C of all installed inverter AC power in the Initial Group, as stated for ICOD in the Summary Sheet of schedule 1;

"Nominal Installed Power Rating" or "PNOM" means the sum of the nameplate rating at 50°C of all installed inverter AC power in the Plant, as stated for Contract Year 1 in the Summary Sheet of schedule 1.

2. Calculation of Deemed Energy

2.1 The following flow chart describes the process to calculate Deemed Net Electrical Energy and refers to corresponding sections of this schedule 17.

[1] Note to Bidders - This is relevant if Bidder's Proposal is based on commissioning the Plant in phases/groups through ICOD structure.

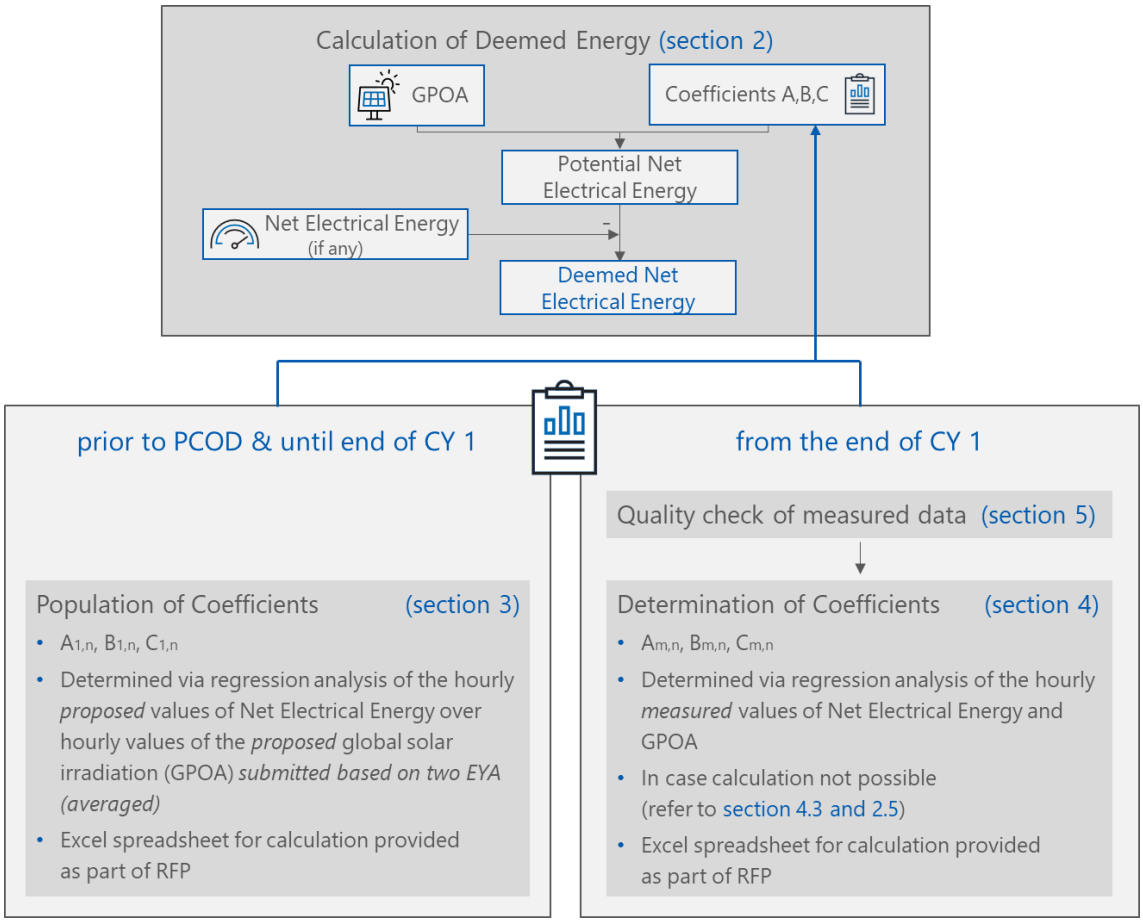

Figure 1: Flow chart illustrating the process to calculate Deemed Net Electrical Energy, referring to corresponding sections of this schedule 17 for further descriptions.

2.2 The Deemed Net Electrical Energy shall be determined for each Deemed Net Electrical Energy Interval (i) within a Billing Period (n) taking into account the actual solar irradiation on plane of array (GPOA) recorded by the meteorological measurement equipment and the applicable Nominal Installed Power Rating or Initial Nominal Installed Power Rating using the formula in the following paragraphs.

2.3 The whole calculation of Deemed Net Electrical Energy in accordance with the following sections shall be represented and verifiable in the Plant Performance Model as described in schedule 1 of the Agreement.

2.4 The Net Electrical Energy as generated by the Plant is a function of the GPOA and shall be expressed as an idealized approximation with a polynomial equation. The equation is the result of a regression analysis that can be generated in spreadsheet software (the "FICHTNER PVc-Tool" attached to this PPA Schedule 17) in a straightforward manner using the equation provided below. Before the end of the first Contract Year, predefined coefficients are applied for the calculation of Deemed Net Electrical Energy (see section 3). For the period from the end of Contract Year 1 the coefficients are calculated based on measured production and irradiation data (see section 4). A quality check of measured data is performed (see section 5).

2.5 The Deemed Net Electrical Energy ($NEE_{n,deemed}$) of the Plant in Billing Period (n) is determined as follows:

$$\mathbf{NEE_{n,deemed} = \sum_{i=1}^{I} NEE_{i,deemed}}$$

where:

$NEE_{n,deemed}$ = Deemed Net Electrical Energy in Billing Period (n) (in kWh)

$NEE_{i,deemed}$ = Deemed Net Electrical Energy in each Deemed Net Electrical Energy Interval (i) (in kWh) in Billing Period (n), as specified below

i = Deemed Net Electrical Energy Interval

I = Number of Deemed Net Electrical Energy Intervals in Billing Period (n)

The Deemed Net Electrical Energy ($NEE_{i,deemed}$) of the Plant in a Deemed Net Electrical Energy Interval (i) is determined as follows:

Case one:
For Deemed Net Electrical Energy Intervals (i) before Project Commercial Operation Date:

$$\mathbf{NEE_{i,deemed} = MIN\ [\ (\ MDL * 1000\ kWh/MW\ )\ ,\ MAX\ [\ 0,\ (\ IPNOM / PNOM\ ) * (\ A_{1,n} * GPOA_i^2 + B_{1,n} * GPOA_i + C_{1,n}\ ) - NEE_{i,act}\ ]\ ]}$$

Case two:
For Deemed Net Electrical Energy Intervals (i) from and after Project Commercial Operation Date:

$$\mathbf{NEE_{i,deemed} = MIN\ [\ (\ MDL * 1000\ kWh/MW\ )\ ,\ MAX\ [\ 0,\ (\ A_{m,n} * GPOA_i^2 + B_{m,n} * GPOA_i + C_{m,n}\ ) - NEE_{i,act}\ ]\ ]}$$

where in all cases:

$NEE_{i,deemed}$ = as previously defined

$NEE_{i,act}$ = Net Electrical Energy (in kWh) of the Plant as actually delivered to the Offtaker in the applicable Deemed Net Electrical Energy Interval (i) and accounted for already according to the procedures set forth in section 2.1 of schedule 9 of the Agreement

IPNOM = Initial Nominal Installed Power Rating (in MW_{AC}) which shall not exceed PNOM as stated in the Summary Sheet of schedule 1

PNOM = Nominal Installed Power Rating (in MW_{AC}) of the entire Plant at Project Commercial Operation Date as stated in the Summary Sheet of schedule 1

MDL = Maximum Dispatch Limit (in MW_{ac}) as defined in Clause 1.1 of the Agreement

$GPOA_i$ = global solar irradiation on Plane of Array (in kWh/m^2) in applicable Deemed Net Electrical Energy Interval (i), determined as the average irradiation measured during the one hour Deemed Net Electrical Energy Interval (i) by on-site pyranometers placed in the plane of array of the PV modules

$A_{1,n}$ = coefficient A in applicable Billing Period (n) of Contract Year 1 as specified below (applicable prior to the Project Commercial Operation Date and from the commencement of the first Contract Year until the end of Contract Year 1)

$B_{1,n}$ = coefficient B in applicable Billing Period (n) of Contract Year 1 as specified below (applicable prior to the Project Commercial Operation Date and from the commencement of the first Contract Year until the end of Contract Year 1)

$C_{1,n}$ = coefficient C in applicable Billing Period (n) of Contract Year 1 as specified below (applicable prior to the Project Commercial Operation Date and from the commencement of the first Contract Year until the end of Contract Year 1)

$A_{m,n}$ = coefficient A in applicable Billing Period (n) of Contract Year (m) as specified below

$B_{m,n}$ = coefficient B in applicable Billing Period (n) of Contract Year (m) as specified below

$C_{m,n}$ = coefficient C in applicable Billing Period (n) of Contract Year (m) as specified below

i = as previously defined

MAX = mathematical function to ensure no "negative" values for Deemed Net Electrical Energy

MIN = mathematical function to ensure no values for Deemed Net Electrical Energy above Maximum Dispatch Limit

2.6 In case that coefficients $A_{m,n}$, $B_{m,n}$ and $C_{m,n}$ could not be determined but were substituted for a Billing Period (n) of Contract Year (m) as described in paragraph 4.3, any Deemed Net Electrical Energy calculated for this particular Billing Period (n) shall be multiplied by the ratio between the Projected Net Electrical Energy for Contract Year (m) and the Projected Net Electrical Energy for Contract Year (m-x) as described in section 4.3 of this schedule 17 considered for the determination of coefficients $A_{m,n}$, $B_{m,n}$ and $C_{m,n}$. Both values of Projected Net Electrical Energy shall be as set forth in the Summary Sheet of schedule 1.

3. Population of coefficients $A_{1,n}$, $B_{1,n}$ and $C_{1,n}$ – prior to the Project Commercial Operation Date and from the commencement of the first Contract Year until the end of Contract Year 1

3.1 The coefficients $A_{1,n}$, $B_{1,n}$ and $C_{1,n}$ of the polynomial equation shall be determined via regression analysis of hourly values of Net Electrical Energy over hourly values of GPOA.

The determination of coefficients $A_{1,n}$, $B_{1,n}$ and $C_{1,n}$ shall be based on the proposed hourly profile of Net Electrical Energy and GPOA data submitted based on two EYA in response to the RFP.

The sum of hourly Net Electrical Energy values shall coincide with the guaranteed Projected Net Electrical Energy of Contract Year 1.

3.2 Any hours with GPOA values for which Net Electrical Energy in any of the hourly time series of the two EYA is

(a) clipped due to the Maximum Dispatch Limit; or
(b) negative

shall be disregarded for determination of the coefficients $A_{1,n}$, $B_{1,n}$ and $C_{1,n}$.

GPOA and Net Electrical Energy values of both EYA shall be averaged for each remaining hour of the year applicable for determination of the coefficients $A_{1,n}$, $B_{1,n}$ and $C_{1,n}$,.

As coefficients $A_{1,n}$, $B_{1,n}$ and $C_{1,n}$ shall vary with the month / Billing Period, their determination shall be performed on a monthly basis considering each remaining hour of the month / Billing Period applicable for determination of the coefficients $A_{1,n}$, $B_{1,n}$ and $C_{1,n}$ in a separate regression analysis.

A spreadsheet tool ("FICHTNER PVc-Tool") is provided as Annex A of PPA Schedule 17 for the population of coefficients $A_{1,n}$, $B_{1,n}$ and $C_{1,n}$.

Coefficients $A_{1,n}$, $B_{1,n}$ and $C_{1,n}$ shall be provided as integer values.

3.3 The determined values for coefficients $A_{1,n}$, $B_{1,n}$ and $C_{1,n}$ vary with the month / Billing Period in which the corresponding Deemed Net Electrical Energy Interval occurs and are set forth in Table 1:

Table 1: Coefficients $A_{1,n}$, $B_{1,n}$ and $C_{1,n}$ prior to the Project Commercial Operation Date and from the commencement of Contract Year 1 until the end of Contract Year 1

(n) Month / Billing Period	$A_{1,n}$	$B_{1,n}$	$C_{1,n}$
(1) January	-7,832.00	109,909.00	-137.00
(2) February	-4,621.00	107,539.00	-101.00
(3) March	-477.00	102,641.00	-95.00
(4) April	-7,400.00	107,418.00	-149.00
(5) May	-14,434.00	111,593.00	-163.00
(6) June	-30,475.00	123,137.00	-355.00

(n) Month / Billing Period	$A_{1,n}$	$B_{1,n}$	$C_{1,n}$
(7) July	-30,829.00	123,800.00	-362.00
(8) August	-28,756.00	119,825.00	-351.00
(9) September	-29,077.00	119,169.00	-368.00
(10) October	-22,523.00	116,126.00	-318.00
(11) November	-26,891.00	117,606.00	-371.00
(12) December	-21,752.00	112,714.00	-328.00

4. Determination of Coefficients – for the period from the end of Contract Year 1

4.1 At the end of every Contract Year (m), the Company shall submit to the Offtaker for Offtaker's approval within seven (7) Days after the end of that Contract Year, its proposed monthly values for the coefficients $A_{m+1,n}$, $B_{m+1,n}$ and $C_{m+1,n}$ to be used in the determination of Deemed Net Electrical Energy for the immediate following Contract Year (m+1), provided

(a) if the Project Commercial Operation Date occurs on or before the 15th Day of a calendar month, then such calendar month is considered in its entirety as the first Billing Period of Contract Year 1 and subsequently of any following Contract Year solely for the purposes of determination of the coefficients A, B and C in accordance with this section 4 of this schedule 17; and provided further

(b) if the Project Commercial Operation Date occurs after the 15th Day of a calendar month, then such calendar month is considered in its entirety as the last Billing Period of Contract Year 1 and subsequently of any following Contract Year solely for the purposes of determination of the coefficients A, B and C in accordance with this section 4 of this schedule 17.

4.2 The proposed monthly values for the coefficients $A_{m+1,n}$, $B_{m+1,n}$ and $C_{m+1,n}$ for the same Billing Period (n) in the immediate following Contract Year (m+1) shall be determined via regression analysis of the hourly measured values of Net Electrical Energy and GPOA of the previous twelve (12) Billing Periods.

Any hours with GPOA values for which Net Electrical Energy is

(a) clipped due to the Maximum Dispatch Limit, or
(b) negative

shall be disregarded for determination of the coefficients $A_{m+1,n}$, $B_{m+1,n}$ and $C_{m+1,n}$.

Coefficients $A_{m+1,n}$, $B_{m+1,n}$ and $C_{m+1,n}$ shall be provided as integer values.

A spreadsheet tool ("FICHTNER PVc-Tool") is provided as Annex A of PPA Schedule 17 for the determination of coefficients $A_{m+1,n}$, $B_{m+1,n}$ and $C_{m+1,n}$.

4.3 Should the determination of coefficients $A_{m+1,n}$, $B_{m+1,n}$ and $C_{m+1,n}$ for a particular Billing Period (n) of Contract Year (m+1) not be possible because for reasons not attributable to the Company, e.g. because Net Electrical Energy was not produced during the corresponding Billing Period (n) of Contract Year (m), then, subject to paragraph 4.4, coefficients $A_{m-x,n}$, $B_{m-x,n}$ and $C_{m-x,n}$ for this particular Billing Period (n) of the previous

Contract Year (m-x) in which Net Electrical Energy was last produced and coefficients were determined shall be applied and substitute the determination of the coefficients $A_{m+1,n}$, $B_{m+1,n}$ and $C_{m+1,n}$ for this particular Billing Period (n) of Contract Year (m+1).

In the event that no previous particular Billing Period (n) in which Net Electrical Energy was produced exists, then coefficients $A_{m+1,n}$, $B_{m+1,n}$ and $C_{m+1,n}$ for the affected Billing Period (n) of Contract Year (m+1) shall be as set forth in Table 1.

Should the Offtaker and the Company be unable to mutually agree if the determination of coefficients $A_{m+1,n}$, $B_{m+1,n}$ and $C_{m+1,n}$ for a particular Billing Period (n) of Contract Year (m+1) is possible or not, the Dispute shall be referred to the Independent Expert for resolution pursuant to Clause 21.2 of the agreement.

4.4 Within seven (7) Days after receipt of each such proposed values for the coefficients $A_{m+1,n}$, $B_{m+1,n}$ and $C_{m+1,n}$, the Offtaker shall either approve such proposed values for the coefficients A, B and C or notify the Company of required revisions to be made to such proposed values for the coefficients $A_{m+1,n}$, $B_{m+1,n}$ and $C_{m+1,n}$. The Company's revised proposed values for the coefficients A, B and C shall be submitted to the Offtaker within seven (7) Days of such notification and the Offtaker shall notify the Company of its approval or disapproval within seven (7) Days of such submittal. The failure of the Offtaker to respond within the applicable period, unless extended by mutual agreement, shall be deemed as approval by the Offtaker of the Company's proposed values for the coefficients $A_{m+1,n}$, $B_{m+1,n}$ and $C_{m+1,n}$ to be used in the determination of Deemed Net Electrical Energy for the immediate following Contract Year (m+1).

5. Quality check of measured data

5.1 Only data from records where all input parameters are valid and within specified limits shall be used. The requirements of IEC 61724-1:2017 shall be followed for all aspects of monitoring and data verification.

5.2 Recorded meteorological and power generation data shall be sampled at no greater than five second intervals and averaged at no greater than five-minute intervals. However, the coefficients shall be determined via regression analysis of hourly measured values of Net Electrical Energy over hourly measured values of the GPOA. For avoidance of doubt, additional losses or degradation shall not be additionally considered in the determination of the coefficients A, B and C as these are already inherent in the measured production data.

5.3 Complete data sets must be submitted by the Company. Averaging of multiple quality checked sensors is required. Offtaker should have the right to review the averaging methodology. Company will submit only one set of data of GPOA which will be the basis for Deemed Net Electrical Energy calculation.

5.4 If one of the input parameters of a data record is marked as invalid (e.g. the data acquisition system reports the readings of all redundant irradiance sensors as out of range) then the data record shall also be marked as invalid.